

H524 Introduction to Biostatistics

Week 3: Probability Distributions and Sampling

Homework Assignment

Fall 2025

Due Date: End of Week 4

Total Points: 100 (+ 5 bonus)

Format: Submit through Canvas

Instructions

- Show all work for full credit
- Round probabilities to 4 decimal places unless otherwise specified
- Round percentages to 1 decimal place
- Include R code where specified
- You may use R to check your work

1 Part 1: Binomial Distribution (25 points)

Question 1: Binomial Conditions (5 points)

A pharmaceutical company claims that their new medication is effective for 80% of patients. Which statement BEST describes why treating 15 patients constitutes a binomial experiment?

- A) Each patient has two outcomes (success/failure), probability stays constant at 0.80, trials are independent, and there are a fixed number (15) of trials
- B) Each patient can have multiple outcomes, the probability changes with each patient, trials depend on previous results, and sample size is variable
- C) The medication works most of the time, some patients get better, the trials happen sequentially, and $n=15$
- D) Each patient is tested once, the success rate is high, patients don't affect each other, and we count total successes

Question 2: Binomial Probability - Exactly 12 Successes (3 points)

For the medication scenario ($n = 15$, $p = 0.80$), what is the probability that exactly 12 patients respond successfully?

- A) 0.2501
- B) 0.8000

- C) 0.1876
- D) 0.3980

Question 3: Binomial Probability - At Least 13 Successes (3 points)

For the same scenario, what is the probability that at least 13 patients respond successfully?

- A) 0.6482
- B) 0.3980
- C) 0.2309
- D) 0.5000

Question 4: R Code for Binomial (2 points)

Which R code correctly calculates $\Pr(X \geq 13)$ for $n = 15$, $p = 0.80$?

- A) `sum(dbinom(13:15, 15, 0.80))` or `pbinom(12, 15, 0.80, lower.tail=FALSE)`
- B) `dbinom(13, 15, 0.80)`
- C) `pbinom(13, 15, 0.80)`
- D) `1 - dbinom(0:12, 15, 0.80)`

Question 5: Binomial Expected Value (2 points)

For the binomial scenario above ($n = 15$, $p = 0.80$), what is the expected number of successful responses?

- A) 12
- B) 15
- C) 10
- D) 9.6

Question 6: Binomial Standard Deviation (2 points)

What is the standard deviation for this binomial distribution?

- A) 1.549
- B) 2.400
- C) 12.000
- D) 0.800

Question 7: Vaccine Efficacy - Lower Tail (3 points)

A vaccine has 90% efficacy. A clinic vaccinates 25 people. What is the probability that fewer than 20 people are protected?

- A) 0.0334
- B) 0.1607
- C) 0.9018
- D) 0.5000

Question 8: Vaccine Efficacy - Range (3 points)

For the same scenario, what is the probability that between 22 and 24 people (inclusive) are protected?

- A) 0.6918
- B) 0.2659
- C) 0.1994
- D) 0.3826

2 Part 2: Normal Distribution (25 points)**Question 9: Z-Score Calculation (2 points)**

Adult heights are normally distributed with mean 170 cm and standard deviation 10 cm. What is the Z-score for a person who is 185 cm tall?

- A) 1.5
- B) 0.15

- C) 2.0
- D) -1.5

Question 10: Upper Tail Probability (3 points)

Using the same height distribution, what proportion of adults are taller than 185 cm?

- A) 0.0668
- B) 0.9332
- C) 0.1587
- D) 0.5000

Question 11: Probability Between Two Values (3 points)

What proportion of adults have heights between 165 cm and 180 cm?

- A) 0.5328
- B) 0.3085
- C) 0.6915
- D) 0.8413

Question 12: 90th Percentile (3 points)

Using the height distribution (mean 170 cm, SD 10 cm), what height represents the 90th percentile?

- A) 182.8 cm
- B) 190.0 cm
- C) 180.0 cm
- D) 175.6 cm

Question 13: 25th Percentile (2 points)

What height represents the 25th percentile?

- A) 163.3 cm
- B) 165.0 cm
- C) 167.5 cm
- D) 160.0 cm

Question 14: Empirical Rule Range (2 points)

Cholesterol levels are normally distributed with mean 200 mg/dL and SD 40 mg/dL. Using the empirical rule, within what range do approximately 95% of cholesterol values fall?

- A) 120 to 280 mg/dL
- B) 160 to 240 mg/dL
- C) 180 to 220 mg/dL
- D) 100 to 300 mg/dL

Question 15: Extreme Cholesterol Values (3 points)

What proportion have cholesterol levels above 280 mg/dL?

- A) 0.0228
- B) 0.9772
- C) 0.0500
- D) 0.1587

Question 16: Hypertension Proportion (2 points)

Systolic blood pressure is normally distributed with mean 125 mmHg and SD 18 mmHg. What proportion have blood pressure above 140 mmHg (hypertension)?

- A) 0.2024
- B) 0.7976
- C) 0.1587
- D) 0.0668

Question 17: Expected Number Hypertensive (2 points)

If 1000 people are screened, approximately how many would be classified as hypertensive?

- A) 202
- B) 798
- C) 159
- D) 140

3 Part 3: Sampling Distributions and CLT (25 points)

Question 18: Sampling Distribution Mean (2 points)

A population has mean $\mu = 100$ and standard deviation $\sigma = 20$. If you take a sample of size $n = 25$, what is the mean of the sampling distribution of $\bar{X}$?

- A) 100
- B) 20
- C) 4
- D) 25

Question 19: Standard Error Calculation (2 points)

What is the standard error of $\bar{X}$ for $n = 25$?

- A) 4
- B) 20
- C) 100
- D) 0.8

Question 20: Effect of Sample Size on SE (3 points)

If $n = 100$ instead of $n = 25$, what is the new standard error?

- A) 2
- B) 4
- C) 10
- D) 1

Question 21: CLT Distribution (2 points)

A population of glucose levels has mean 100 mg/dL and standard deviation 25 mg/dL. A sample of 64 individuals is randomly selected. According to the Central Limit Theorem, the sample mean $\bar{X}$ is distributed as:

- A) $N(100, 3.125^2)$
- B) $N(100, 25^2)$

- C) $N(100, 0.391^2)$
- D) $N(64, 25^2)$

Question 22: Sample Mean Probability - Upper Tail (4 points)

For the glucose scenario above, what is the probability that the sample mean exceeds 105 mg/dL?

- A) 0.0548
- B) 0.4207
- C) 0.9452
- D) 0.5000

Question 23: Sample Mean Probability - Range (2 points)

What is the probability that the sample mean glucose level is between 97 and 103 mg/dL?

- A) 0.6629
- B) 0.3830
- C) 0.4515
- D) 0.7881

Question 24: SD vs SE Calculation (2 points)

If the population standard deviation is 30 and sample size is 36, what is the standard error?

- A) 5
- B) 30
- C) 6
- D) 1.2

Question 25: Effect of Sample Size on SE (2 points)

As sample size increases, what happens to the standard error and why is this important?

- A) SE decreases; this makes our estimates more precise and reliable for inference
- B) SE increases; this makes our estimates more conservative
- C) SE stays the same; sample size doesn't affect variability
- D) SE decreases; this makes the population standard deviation smaller

Question 26: SD vs SE Concept (1 point)

What is the key difference between standard deviation (SD) and standard error (SE)?

- A) SD measures variability of individual observations; SE measures variability of sample means
- B) SD measures variability of sample means; SE measures variability of individual observations
- C) SD is always larger than SE
- D) SD and SE are the same when $n=1$

Question 27: CLT Birth Weight Probability (3 points)

Birth weights have mean 3400g and SD 500g. You sample 100 births. What is the probability that the sample mean birth weight is less than 3350g?

- A) 0.1587
- B) 0.4602
- C) 0.8413
- D) 0.0228

Question 28: R Code for CLT (2 points)

Which R code correctly calculates the probability in Question 27?

- A) `pnorm(3350, 3400, 500/sqrt(100))` or `pnorm(3350, 3400, 50)`
- B) `pnorm(3350, 3400, 500)`
- C) `pbinom(3350, 100, 0.5)`
- D) `pnorm(3350, 3400, 50, lower.tail=FALSE)`

4 Part 4: Confidence Intervals (20 points)

Question 13: Manual CI Calculation (10 points)

A study measures pain reduction scores for 20 patients receiving a new treatment:

Sample mean: $\bar{x} = 6.2$ points

Sample standard deviation: $s = 2.4$ points

Sample size: $n = 20$

- a) Should you use a z or t distribution? Why? (2 points)
- b) Find the appropriate critical value for a 95% confidence interval. (2 points)
- c) Calculate the standard error. (2 points)
- d) Calculate the margin of error. (2 points)
- e) Construct the 95% confidence interval. (2 points)

Question 29: CI Interpretation (2 points)

Based on Question 13, which statement correctly interprets the 95% confidence interval for mean pain reduction?

- A) We are 95% confident that the true mean pain reduction for all patients receiving this treatment is between 5.08 and 7.32 points
- B) 95% of patients will have pain reduction between 5.08 and 7.32 points
- C) There is a 95% probability that the sample mean is between 5.08 and 7.32 points
- D) The population mean is definitely between 5.08 and 7.32 points with 95% certainty

Question 30: Meaning of 95% Confidence (2 points)

What does "95% confident" mean in the context of confidence intervals?

- A) If we repeated this study many times, 95% of the confidence intervals we construct would contain the true population mean
- B) 95% of the data values fall within the confidence interval
- C) There is a 95% probability that our specific interval contains the population mean
- D) We are 95% sure our sample mean equals the population mean

Question 31: Comparing Confidence Levels (1 point)

If you calculated a 99% confidence interval instead of 95%, the interval would be:

- A) Wider, because we need a larger range to be more confident we've captured the true mean
- B) Narrower, because we are more confident about the estimate
- C) The same width, confidence level doesn't affect interval width
- D) Wider, but only if sample size increases

Question 15: CI with R (5 points)

A clinical trial measures cholesterol reduction for 30 patients:

```
cholesterol_reduction <- c(28, 32, 25, 30, 27, 35, 22, 29, 31, 26,  
24, 33, 28, 30, 27, 29, 31, 25, 28, 32,  
26, 30, 29, 27, 34, 28, 31, 25, 29, 30)
```

- a) Write R code to calculate a 95% confidence interval using `t.test()`. (2 points)
- b) Report the confidence interval. (1 point)
- c) Based on this interval, is there strong evidence that the mean reduction exceeds 25 mg/dL? Explain. (2 points)

5 Part 5: Integration and R Applications (5 points)

Question 16: Comprehensive Problem (5 points)

A pharmaceutical company tests a new drug. In trials with 80 patients:

- 72 patients respond successfully
 - Mean cholesterol reduction: $\bar{x} = 35$ mg/dL
 - Standard deviation: $s = 8$ mg/dL
- a) Calculate a 95% confidence interval for the true success rate (proportion). Hint: For proportions, use $\hat{p} \pm 1.96\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ (2 points)
 - b) Calculate a 95% confidence interval for the true mean cholesterol reduction. (2 points)
 - c) Write complete R code to perform both calculations and create a visualization of the cholesterol reduction data. (1 point)

Bonus: AI-Enhanced Learning (5 bonus points)

Question 17: AI Verification and Learning (5 bonus points)

Choose ONE problem from Part 3 (Sampling Distributions and CLT).

- a) Ask an AI tool (Claude, ChatGPT, etc.) to solve the problem. Include your exact prompt. (1 point)
- b) Verify the AI's answer by working through it yourself step-by-step. Did the AI get it correct? If not, identify the error. (2 points)
- c) Ask the AI to explain the Central Limit Theorem using a medical example different from those in class. Evaluate the quality of the explanation. (2 points)

Submission Requirements

Required Files:

1. **PDF Document** with all written work, calculations, and interpretations
2. **R Script File (.R)** with all R code properly commented
3. **AI Usage Documentation** (if applicable for bonus):
 - Prompts used
 - AI responses
 - Your verification and evaluation

Formatting Guidelines:

- Clearly label each question number
- Show all calculation steps for full credit
- For probability problems, write both the formula and the numerical answer
- Round consistently per instructions (probabilities to 4 decimals, percentages to 1 decimal)
- Include units where appropriate (mg/dL, mmHg, cm, etc.)

Grading Rubric

Category	Points	Criteria
Calculations	40	Correct formulas, accurate arithmetic
Methods	30	Appropriate approach, proper formulas
Interpretation	20	Clear explanations, correct context
R Code	10	Working code, proper syntax, comments
Bonus	+5	Thoughtful AI integration and evaluation

Common Mistakes to Avoid

- **Binomial:** Forgetting to check conditions; using continuous formulas for discrete distributions
- **Normal:** Confusing $\Pr(X > a)$ with $\Pr(X < a)$; forgetting to standardize
- **Sampling Distributions:** Using population SD instead of SE; confusing individual vs. sample mean distributions
- **Confidence Intervals:** Using z when you should use t ; misinterpreting what "95% confident" means
- **R Code:** Using `lower.tail=TRUE` when you need `lower.tail=FALSE`; forgetting to specify `mean` and `sd` parameters

Key Formulas

Binomial Distribution:

$$\Pr(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad E(X) = np, \quad \text{SD}(X) = \sqrt{np(1-p)}$$

Standardization:

$$Z = \frac{X - \mu}{\sigma}$$

Sampling Distribution:

$$E(\bar{X}) = \mu, \quad \text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

Confidence Interval (t-distribution):

$$\bar{x} \pm t_{\alpha/2, n-1} \times \frac{s}{\sqrt{n}}$$

Academic Integrity

AI Usage Guidelines

You are encouraged to:

- + Use AI tools to check your work
- + Ask AI to explain concepts you don't understand
- + Verify your calculations with AI
- + Use AI to help debug R code

You must NOT:

- Submit AI-generated work without understanding it
- Copy solutions without showing your own reasoning
- Use AI as a substitute for learning the material
- Rely on AI for exam preparation without independent practice

Remember: The goal is to develop YOUR ability to work with probability distributions and understand statistical inference. AI is a tool to enhance learning, not replace it.

Study Tips

1. **Practice with R:** Work through all the lab exercises before attempting this assignment
2. **Verify by hand:** For simple problems, calculate by hand first, then check with R
3. **Understand, don't memorize:** Focus on understanding when to use each distribution

4. **Draw pictures:** Sketch normal curves and shade regions for probability problems
5. **Check your answers:** Probabilities must be between 0 and 1; CIs should make sense in context

Resources

- **Textbook:** Pagano & Gauvreau, Chapters 7 and 8
- **Lecture Slides:** Week 3 slides with examples
- **Lab Materials:** Week 3 lab with step-by-step R code
- **R Documentation:** `?dbinom`, `?pnorm`, `?t.test`
- **Office Hours:** Wednesday 10:00-11:30am or by appointment